PROJECTED MULTILINEAR TREES: EXACT ERROR DECOMPOSITION, UNIVERSAL BOUNDS, AND INDEPENDENT-LAW SHARPNESS

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ABSTRACT. We study finite typed trees whose internal vertices apply multilinear laws and whose recursively evaluated outputs are orthogonally projected onto reduced subspaces. An exact child-error expansion separates local closure leakage from propagated and higher-order branch interactions. This yields the universal ambient and projected-root coefficients k and $k-1$. We then close a fixed-parameter sharpness question that was previously unresolved: for real binary two-node chains with independent bilinear laws, rank-one orthogonal projection, unit leaves, and $\rho = \eta M$, the projected coefficient is exactly sharp, $C^P_{2,\mathrm{ind}}(\eta) = 1$ for $0 < \eta \le 1$. The earlier η^2 family is retained as a restricted repeated-law construction, not as evidence against the general theorem. For $k = 3$ we provide certified independent-law lower witnesses for chain and branching topologies while leaving global sharpness open.

1. Scope and epistemic status

This paper is the mathematical core of a larger repository. Executable behavior is governed by the source and tests; mathematical status is governed by the theorem registries and proof dossiers. Numerical residuals and finite checks are reported as evidence, not promoted to universal theorems. The finite computational core is frozen at commit `1f4984ec8e741049789e0035c7a3ba84c86d3f29`; the v5 theorem package is recorded at the subsequent scientific commits listed in the reproducibility companion.

2. Typed projected trees

Let $\mathcal{T}$ be a finite rooted tree. For every type τ, let V_τ and W_τ be finite-dimensional real or complex Hilbert spaces and let $Q_\tau : W_\tau \to V_\tau$ be an isometry. Put

$$P_\tau = Q_\tau Q_\tau^*, \qquad P_\tau^2 = P_\tau = P_\tau^*.$$

An internal vertex v of arity a_v carries a bounded typed law

$$\mu_v : \prod_{j=1}^{a_v} V_{\tau(v,j)} \longrightarrow V_{\tau(v)}.$$

Write $M_v = \|\mu_v\|_{\mathrm{op}}$ and define the local closure map

$$r_v = (I - P_v)\mu_v(P_{c_1}\cdot, \dots, P_{c_{a_v}}\cdot), \qquad \rho_v = \|r_v\|_{\mathrm{op}}.$$

For reduced leaves $z_\ell \in W_{\tau(\ell)}$, define the ambient and recursively projected evaluations by

$$F_\ell = R_\ell = Q_{\tau(\ell)} z_\ell, \quad F_v = \mu_v(F_{c_1}, \dots, F_{c_{a_v}}), \quad R_v = P_v \mu_v(R_{c_1}, \dots, R_{c_{a_v}}).$$

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Research manuscript. The theorem statuses in this paper are scoped to the stated hypotheses. Independent human review and theorem-level novelty review remain pending.

At the root, define

$$E^A = \|F - R\|, \quad E^P = \|PF - R\|, \quad E^N = \|(I-P)F\|, \quad E^{\mathrm{red}} = \|Q^*F - Q^*R\|.$$

Proposition 2.1 (Exact root geometry)**.** *For every valid typed tree,*

$$(E^A)^2 = (E^P)^2 + (E^N)^2, \qquad E^{\mathrm{red}} = E^P.$$

The first identity is the orthogonal decomposition $F - R = (PF - R) + (I - P)F$; the second follows from the isometry of Q on the projected range.

3. Exact local error algebra

Let $D_v = F_v - R_v$, and for a nonempty subset S of child slots let $y_i^S = D_i$ when $i \in S$ and $y_i^S = R_i$ otherwise. Multilinearity gives the exact identity

$$D_v = r_v(R_1, \ldots, R_{a_v}) + \sum_{\varnothing \neq S \subseteq [a_v]} \mu_v(y_1^S, \ldots, y_{a_v}^S). \tag{1}$$

Applying P_v annihilates the new local normal residual because $P_v(I - P_v) = 0$. Equation (1) is an identity, not a bound; the inequalities begin only after taking norms and applying operator estimates.

4. Universal coefficients

Let k be the number of internal vertices and $L = \prod_\ell \|z_\ell\|$. Under $M_v \leq M$ and $\rho_v \leq \rho$, induction on the tree gives

Theorem 4.1 (Universal ambient and projected bounds)**.**

$$E^A \leq k\rho M^{k-1}L, \qquad E^N \leq k\rho M^{k-1}L, \qquad E^P = E^{\mathrm{red}} \leq (k-1)\rho M^{k-1}L.$$

The root source contributes to ambient and normal error but not to the projected-root error. This is the exact origin of the coefficient change $k \to k - 1$. The theorem is conditional only on the finite-dimensional typed Hilbert-space, orthogonal-projector, multilinearity, and uniform norm/defect hypotheses stated above.

5. Fixed-eta sharpness for two-node chains

For a fixed binary chain define the normalized projected constant

$$C_T^P(\eta) = \sup \frac{E^P}{\rho M L}, \qquad \eta = \rho/M.$$

The supremum is over the declared admissible class. The universal theorem implies $C_T^P(\eta) \leq 1$ for $k = 2$.

Theorem 5.1 (Sharpness for independent binary laws)**.** *Consider real two-dimensional spaces, the rank-one projector $P = \mathrm{diag}(1, 0)$, unit leaves e_0, e_0, e_0, and independently chosen bilinear laws. For every $0 < \eta \leq 1$, there are laws with $\|\mu_1\|_{\mathrm{op}} = \|\mu_2\|_{\mathrm{op}} = M$, closure defects at most $\rho = \eta M$, and*

$$E^P = \rho M.$$

Consequently,

$$\boxed{C^P_{2,\mathrm{ind}}(\eta) = 1}, \qquad 0 < \eta \leq 1.$$

TABLE 1. V5 sharpness status.

class	result	status
$k = 2$, independent laws	$C^P_{2,\mathrm{ind}}(\eta) = 1$	exact under assumptions
$k = 2$, repeated law	$\eta \le C^P_{2,\mathrm{rep}}(\eta) \le 1$	open
$k = 3$, independent chain	stated analytic lower bound	certified lower bound
$k = 3$, independent branching	stated analytic lower bound	certified lower bound
finite-tree independent conjecture	$C^P_{T,\mathrm{ind}}(\eta) = k(T) - 1$	open conjecture

Proof. Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. Define

$$\mu_1(x, y) = Mx_1y_1e_0 + \rho x_0y_0e_1, \qquad \mu_2(x, y) = Mx_1y_0e_0.$$

For unit x, y, the two output components of μ_1 have squared norm at most $M^2(x_1^2y_1^2+\eta^2x_0^2y_0^2) \le M^2$; equality is attained on the normal-normal input. The second law has norm M and the same estimate is immediate. On projected inputs, the first closure map has norm ρ and the second has norm zero. The ambient first-node value on (e_0, e_0) is ρe_1, while its projected value is zero. The second law maps this normal error with gain M into $\operatorname{ran} P$. Hence $E^P = M\rho$. The upper bound in Theorem 4.1 supplies the matching inequality. □

Remark 5.2 (Parameter sharing changes the extremal problem). The earlier gated-planar/repeated-law construction gives $E^P = \eta^2M^2$, hence only the lower bound $C^P_{2,\mathrm{rep}}(\eta) \ge \eta$ after normalization. The universal upper remains $C^P_{2,\mathrm{rep}}(\eta) \le 1$. Thus independent-law and repeated-law sharpness are distinct extremal problems.

6. Independent-law k=3 lower witnesses

The v5 construction package supplies separate chain and branching witnesses. With $q = \min(\rho, M/\sqrt{2})$, both yield

$$E^P = 2Mq\sqrt{M^2 - q^2}.$$

Therefore, relative to the $k = 3$ normalization ρM^2,

$$C^P_{3,\mathrm{ind}}(\eta) \ge \begin{cases} 2\sqrt{1-\eta^2}, & 0 < \eta \le 1/\sqrt{2}, \\ 1/\eta, & 1/\sqrt{2} < \eta \le 1. \end{cases}$$

These are certified construction lower bounds. They do not close the global constant $C^P_T(\eta) \le 2$, and equality remains open for both topologies.

7. Nodewise and topology-aware directions

The exact subset expansion supports mixed-mask dynamic programming, source path sums, and scalar telescoping-order optimization. These certificates attribute responsibility but do not imply a universal topology-only sharp constant. Chain, branching, depth, fan-out, fan-in, and operator-sharing patterns must be treated as separate coordinates of the extremal problem. The v5 conjecture that sufficiently free independent laws attain $k-1$ for all finite trees is recorded as open rather than silently inferred from the $k = 2$ theorem.

8. Limitations and conclusion

The results are finite-dimensional and scoped. Repeated-law sharpness, dimension/rank reduction, global $k = 3$ sharpness, globally tight multilinear norms, theorem-level novelty, and independent human review remain open. The paper therefore reports a closed theorem for one natural class, certified lower constructions for the next case, and a precise map of the unresolved extremal frontier.

References

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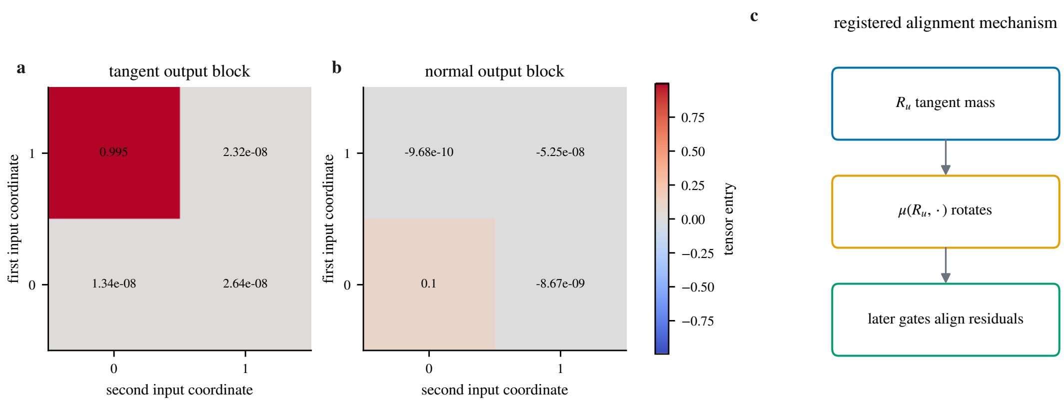

FIGURE 1. Registered projected-tree extremizer geometry reused from the finite v4 evidence package.